

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING (FTE)
Chandubhai S. Patel Institute of Technology (CSPIT)
Department of Computer Science & Engineering

ACADEMIC YEAR: 2025-26

Practical List

Subject: BLOCKCHAIN AND ITS APPLICATIONS
(OCCE3004) (6th Semester)

Exp. No.	Name of Experiment	Hours
1.	Study Blockchain Architecture and Block structure. Perform following tasks: I. Create Blockchain which consists of 5 blocks. [First Block is genesis block]. II. Validate the blocks by implementing mining logic [Cryptographically Mathematical Puzzles] III. Set the difficulty of blocks. IV. Keep timestamp, block version, data and previous hash as block parameters. V. Verify the blocks. VI. Access all transaction. VII. Modify the block data. Implement above all functionalities using Node js/Python/Java/C++/go lang.	2
2.	Install and configure the following development setup tools to implement Blockchain development. o Metamask (Wallet) o Ganache Local Private Blockchain Network o Go-Ethereum (Geth) Client o Truffle framework o Hardhat framework Study and configure all testnets available in Metamask and also setup custom network using Ganache.	2
3.	Configure Geth (Go-Ethereum) over Windows/Linux/Mac operating system. Perform the following tasks: o Build Your Own Ethereum Private Blockchain with two peer nodes locally. o Build Your Own Ethereum Private Blockchain with two peer nodes as	4

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	geographically distributed. o Configure Ethereum testnet such as Sepolia, Goerli and Layer 2.	
4.	Implement basic of Solidity programming: Syntax, Variables, Functions, mapping, access modifiers using Remix online IDE. Use online REMIX IDE by Ethereum to run solidity code.	6
5.	Write a Smart-contract of EVENT PARTICIPATION CERTIFICATE which includes the following things - o Write code using 0.8.0 or higher version of solidity. - o Define a struct to define different properties of certificate issues. - o Define mapping function to struct - o Create a constructor to initialize the first transaction using admin only. - o Create function to issue a new certificate with certificate_id, student_name, branch, course_name and grade - o Initialize proper datatypes of all variable. - o Display the certificate details by certificate_id - o Verify the certificate hash. Use online Ethereum REMIX IDE to compile and deploy the contract. Integrate the metamask wallet to perform the transaction of certificate.	2
6.	Write a Smartcontract to store details of PATIENTS's details which includes the following things - o Write code using 0.8.0 or higher version of solidity. - o Define struct to define different properties of adding patient's details. - o Define mapping function to struct - o Create a constructor to initialize first transaction using admin only. - o Create function to add details of patient such as patient_id, patient_name, decease type, doctor_name and patient_contact. - o Initialize proper datatypes of all variable. - o Display the patient's details by patient_id - o Display all patient's details Use online Ethereum REMIX IDE to compile and deploy the contract. Integrate the metamask wallet to perform the transaction of certificate.	2
7.	Refer to Practical 5: Use Truffle to compile and deploy the contract on Ganache	2

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	and Integrate the Metamask to perform transactions. Test the smart contract using the Mocha and Chai framework before deployment.	
8.	Refer to Practical 6: Use Hardhat to compile and deploy the contract using INFURA API and Integrate the Metamask to perform transactions. Test the smart contract using the Mocha and Chai framework before deployment.	2
9.	Create Dapp (Decentralized Application) and link your client-side application with Blockchain network created in practical 5 th and 6 th .	2
10.	Install and configure the Hyperledger Fabric using the Linux platform. Create permissioned network and implement chain code to transfer the assets from owner to buyer.	2
11.	Create a CHARUSAT Crypto Coin and transfer the initial coin from Admin to different accounts.	2
12.	Create Non-Fungible-Token (NFT) for CHARUSAT Event. NFTs contain meta information such as date, original owner, and offers.	2